

Euler's number



Outline

Euler's number e (Napier's constant)

Euler's number in finance

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Introduction: Euler's number e (Napier's constant)

- ▶ In this section we prove that the sequence $\left(1 + \frac{1}{n}\right)^n$ is **convergent**
- ▶ Its limit is **Euler's number** $e = 2.71828\dots$, one of the most important numbers in mathematics
- ▶ We extend the result to arbitrary divergent sequences and use it to compute many limits of the form 1^∞

Theorem 1

The sequence

$$a_n = \left(1 + \frac{1}{n}\right)^n \quad \text{with } n \geq 1$$

is convergent.

Proof (1/4)

We will prove that the sequence $\{a_n\}$ is non-decreasing ($a_n \geq a_{n-1}, \forall n \in \mathbb{N}, n \geq 2$) and bounded ($m \leq a_n \leq M, \forall n \in \mathbb{N}, n \geq 1$); hence it is convergent by the monotone sequence theorem.

The proof is split into three parts:

- (1) the monotonicity of $\{a_n\}$ and its lower bound
- (2) the monotonicity and the upper bound of an auxiliary sequence $\{b_n\}$
- (3) the synthesis of the two results

Proof (2/4)

Part (1). To prove that $\{a_n\}$ is non-decreasing, we study, for $n \geq 2$, the ratio:

$$\begin{aligned}\frac{a_n}{a_{n-1}} &= \frac{\left(1 + \frac{1}{n}\right)^n}{\left(1 + \frac{1}{n-1}\right)^{n-1}} = \frac{\left(\frac{n+1}{n}\right)^n}{\left(\frac{n}{n-1}\right)^{n-1}} \\&= \left(\frac{\frac{n+1}{n}}{\frac{n}{n-1}}\right)^n \frac{1}{\left(\frac{n}{n-1}\right)^{-1}} = \left(\frac{n^2 - 1}{n^2}\right)^n \frac{1}{\left(\frac{n-1}{n}\right)} \\&= \frac{\left(1 - \frac{1}{n^2}\right)^n}{1 - \frac{1}{n}} \geq \frac{1 - n \cdot \frac{1}{n^2}}{1 - \frac{1}{n}} = 1\end{aligned}$$

Proof (3/4)

where for the “ $\geq$ ” we applied Bernoulli's inequality:

$$(1+x)^n \geq 1 + nx \quad \text{with} \quad x = -\frac{1}{n^2} \geq -1 \quad \text{and} \quad n \geq 2$$

Hence we have

$$\frac{a_n}{a_{n-1}} \geq 1$$

i.e., $a_n \geq a_{n-1}$ and the sequence is non-decreasing.

Proof (4/4)

To prove that $\{a_n\}$ is bounded *from below*, we observe that, since $a_1 = 2$ and the sequence is non-decreasing, it follows that $a_n \geq 2, \forall n \geq 1$. □

Proof (1/4)

Part (2). Consider the auxiliary sequence

$$b_n = \left(1 + \frac{1}{n}\right)^{n+1} \quad \text{with } n \geq 1$$

and let us prove that it is decreasing and bounded from above.

Proof (2/4)

To prove that $\{b_n\}$ is decreasing, we study, for $n \geq 2$, the ratio:

$$\begin{aligned}\frac{b_n}{b_{n-1}} &= \frac{\left(1 + \frac{1}{n}\right)^{n+1}}{\left(1 + \frac{1}{n-1}\right)^n} = \frac{\left(\frac{n+1}{n}\right)^{n+1}}{\left(\frac{n}{n-1}\right)^n} \\&= \left(\frac{\frac{n+1}{n}}{\frac{n}{n-1}}\right)^n \left(\frac{n+1}{n}\right) = \frac{1}{\left(\frac{n^2}{n^2-1}\right)^n} \left(\frac{n+1}{n}\right) \\&= \frac{1}{\left(1 + \frac{1}{n^2-1}\right)^n} \left(\frac{n+1}{n}\right) \leq \frac{1}{\left(1 + \frac{n}{n^2-1}\right)} \left(\frac{n+1}{n}\right) \\&< \frac{1}{\left(1 + \frac{1}{n}\right)} \left(1 + \frac{1}{n}\right) = 1\end{aligned}$$

Proof (3/4)

where for the “ $\leq$ ” we applied Bernoulli's inequality:

$$(1+x)^n \geq 1 + nx \quad \text{with} \quad x = \frac{1}{n^2-1} \geq -1 \quad \text{and} \quad n \geq 2$$

and for the “ $<$ ” we applied the inequality

$$\frac{n}{n^2-1} > \frac{1}{n} \quad \text{since} \quad n^2 > n^2-1 \quad \text{for} \quad n \geq 2$$

Hence we have proved that

Proof (4/4)

$$\frac{b_n}{b_{n-1}} < 1 \quad \text{hence} \quad b_n < b_{n-1}$$

and the sequence $\{b_n\}$ is decreasing.

Since $b_1 = 4$ and the sequence is decreasing, we therefore obtain

$$b_n \leq b_1 = 4, \quad \forall n \geq 1$$

that is, $\{b_n\}$ is bounded from above. □

Proof

Part (3). The two sequences are related by

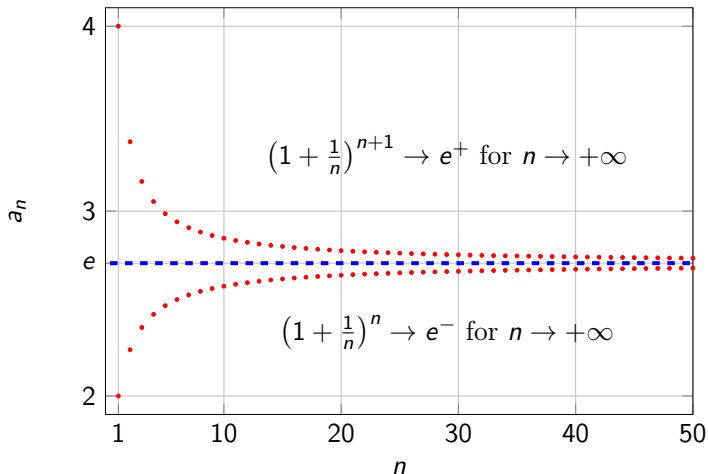
$$b_n = a_n \underbrace{\left(1 + \frac{1}{n}\right)}_{>1, \forall n \geq 1} \quad \text{therefore } b_n > a_n, \forall n \in \mathbb{N}, n \geq 1$$

and consequently

$$2 \leq a_n < b_n \leq 4, \forall n \geq 1$$

The sequence $\{a_n\}$ is therefore non-decreasing and bounded and, by the monotone sequence theorem, it is convergent. □

Example: Graphs of the sequences $\{a_n\}$ and $\{b_n\}$



Euler's number e (Napier's constant)

- ▶ The limit of the sequence a_n just studied is an **irrational** number that is very important in mathematics. This limit is denoted by the letter e (**Euler's number**, or Napier's constant) and its decimal representation begins as follows:

2.7182818284...

By definition, we have:

$$e = \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n \quad (1)$$

- ▶ This number is very often used as the base of logarithms, which, when this base is used, are called natural or Napierian logarithms (after the mathematician **John Napier**) and are denoted simply by the symbol $\log$ (or $\ln$) without indicating the base.
- ▶ Consequently, we have:

$$\lim_{n \rightarrow +\infty} n \log \left(1 + \frac{1}{n} \right) = \lim_{n \rightarrow +\infty} \log \left(1 + \frac{1}{n} \right)^n = \log e = 1$$

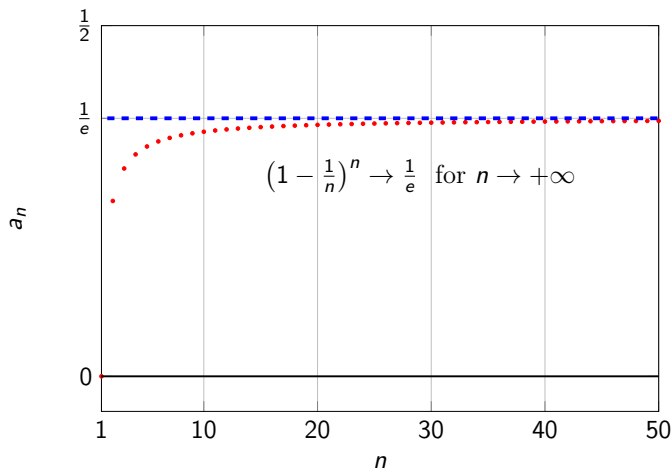
Example: Computing limits with the sequence tending to e ($1/2$)

$$\lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}$$

Since:

$$\left(\frac{n-1}{n}\right)^n = \frac{1}{\left(\frac{n}{n-1}\right)^n} = \frac{1}{\left(1 + \frac{1}{n-1}\right)^n} = \frac{1}{\underbrace{\left(1 + \frac{1}{n-1}\right)^{n-1}}_{\rightarrow e}} \cdot \frac{1}{\underbrace{\left(1 + \frac{1}{n-1}\right)}_{\rightarrow 1}}$$

Example: Computing limits with the sequence tending to e (2/2)



Theorem 2

Let $\{c_n\}$ be any divergent sequence (to $+\infty$ or $-\infty$); then

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{c_n}\right)^{c_n} = e, \quad \lim_{n \rightarrow +\infty} \left(1 - \frac{1}{c_n}\right)^{c_n} = \frac{1}{e} \quad (2)$$

Proof (1/3)

Recall that, given $x \in \mathbb{R}$, its integer part, denoted by $\lfloor x \rfloor$, is the largest integer not exceeding x .

If $c_n \rightarrow +\infty$, since $\lfloor c_n \rfloor > c_n - 1$, by comparison we also have $\lfloor c_n \rfloor \rightarrow +\infty$. Hence, by (1) and by the definition of limit we have

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{\lfloor c_n \rfloor + 1} \right)^{\lfloor c_n \rfloor + 1} = \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{\lfloor c_n \rfloor} \right)^{\lfloor c_n \rfloor} = e$$

Proof (2/3)

Using:

$$\lfloor c_n \rfloor \leq c_n < \lfloor c_n \rfloor + 1$$

we obtain

$$\left(1 + \frac{1}{c_n}\right)^{c_n} < \left(1 + \frac{1}{\lfloor c_n \rfloor}\right)^{\lfloor c_n \rfloor + 1} = \underbrace{\left(1 + \frac{1}{\lfloor c_n \rfloor}\right)^{\lfloor c_n \rfloor}}_{\rightarrow e} \cdot \underbrace{\left(1 + \frac{1}{\lfloor c_n \rfloor}\right)}_{\rightarrow 1}$$

and also

$$\left(1 + \frac{1}{c_n}\right)^{c_n} > \left(1 + \frac{1}{\lfloor c_n \rfloor + 1}\right)^{\lfloor c_n \rfloor} = \underbrace{\left(1 + \frac{1}{\lfloor c_n \rfloor + 1}\right)^{\lfloor c_n \rfloor + 1}}_{\rightarrow e} \cdot \underbrace{\left(1 + \frac{1}{\lfloor c_n \rfloor + 1}\right)^{-1}}_{\rightarrow 1}$$

Proof (3/3)

hence the first limit of the theorem follows from the comparison theorem.

If instead $c_n \rightarrow -\infty$, then the sequence $d_n = -c_n \rightarrow +\infty$, and we have

$$\left(1 + \frac{1}{c_n}\right)^{c_n} = \left(1 - \frac{1}{d_n}\right)^{-d_n} = \left(\frac{d_n}{d_n - 1}\right)^{d_n} = \left(1 + \frac{1}{d_n - 1}\right)^{d_n - 1} \cdot \left(1 + \frac{1}{d_n - 1}\right)$$

Since $d_n - 1 \rightarrow +\infty$, the claim follows from the previous case. The second limit of the theorem is proved analogously.



Euler's number e (Napier's constant)

This theorem is useful for computing limits involving the indeterminate form 1^∞

Example: Computing limits with the sequence tending to e ($1/4$)

We compute

$$\lim_{n \rightarrow +\infty} \left(\frac{n}{3+n} \right)^{5n+1} = [1^\infty]$$

since:

$$\lim_{n \rightarrow +\infty} \frac{n}{3+n} = \lim_{n \rightarrow +\infty} 1 - \frac{3}{3+n} = 1, \quad \lim_{n \rightarrow +\infty} 5n+1 = +\infty$$

We rewrite the sequence as follows:

Example: Computing limits with the sequence tending to e (2/4)

$$\begin{aligned}\left(\frac{n}{3+n}\right)^{5n+1} &= \left(\frac{3+n}{n}\right)^{-(5n+1)} = \frac{1}{\left(1 + \frac{3}{n}\right)^{5n+1}} \\ &= \frac{1}{\left(\left(1 + \frac{3}{n}\right)^{\frac{n}{3}}\right)^{\frac{3(5n+1)}{n}}} = \frac{1}{\left(\left(1 + \frac{1}{\frac{n}{3}}\right)^{\frac{n}{3}}\right)^{\frac{3(5n+1)}{n}}}\end{aligned}$$

Example: Computing limits with the sequence tending to e ($3/4$)

and we consider the denominator

$$\left(\underbrace{\left(1 + \frac{1}{\frac{n}{3}} \right)^{\frac{n}{3}}}_{\rightarrow e} \right)^{\overbrace{3(5n+1)}^{\rightarrow 15}} \rightarrow e^{15} \quad \text{since} \quad \frac{n}{3} \rightarrow +\infty \text{ for } n \rightarrow +\infty$$

Summarizing, we have:

Example: Computing limits with the sequence tending to e ($4/4$)

$$\lim_{n \rightarrow +\infty} \left(\frac{n}{3+n} \right)^{5n+1} = \frac{1}{e^{15}}$$

Alternative method:

$$\lim_{n \rightarrow +\infty} \left(\frac{n}{3+n} \right)^{5n+1} = \lim_{n \rightarrow +\infty} \left(1 - \frac{3}{3+n} \right)^{5n+1} = \lim_{n \rightarrow +\infty} \left(\underbrace{\left(1 - \frac{1}{\frac{3+n}{3}} \right)}_{\rightarrow \frac{1}{e}}^{\frac{3+n}{3}} \right)^{\overbrace{3(5n+1)}^{\rightarrow 15}} = \frac{1}{e^{15}}$$

since

$$\frac{3+n}{3} \rightarrow +\infty \text{ for } n \rightarrow +\infty$$

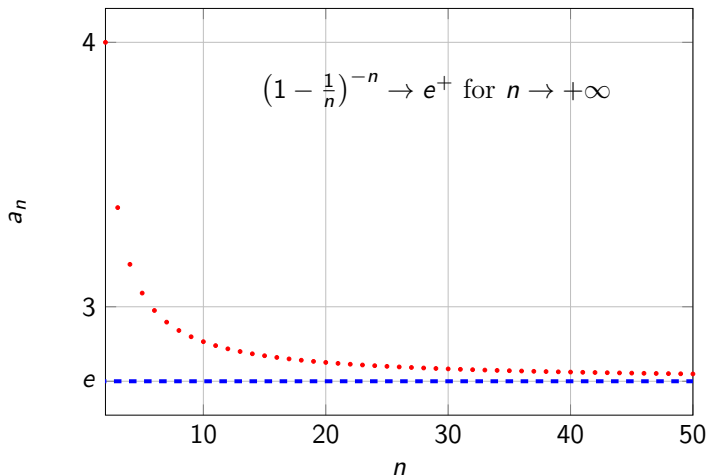
Example: Computing limits with the sequence tending to e ($1/2$)

$$\lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n}\right)^{-n} = e$$

Since:

$$\left(\frac{n-1}{n}\right)^{-n} = \left(\frac{n}{n-1}\right)^n = \left(1 + \frac{1}{n-1}\right)^n = \underbrace{\left(1 + \frac{1}{n-1}\right)^{n-1}}_{\rightarrow e} \cdot \underbrace{\left(1 + \frac{1}{n-1}\right)}_{\rightarrow 1}$$

Example: Computing limits with the sequence tending to e (2/2)



Example: Computing limits with the sequence tending to e (1/3)

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{\alpha}{n}\right)^n = e^\alpha \quad \text{with } \alpha \in \mathbb{R}$$

If $\alpha = 0$ the result is immediate, since the sequence is constant:

$$\left(1 + \frac{0}{n}\right)^n = 1^n = 1 = e^0$$

If instead $\alpha \neq 0$ (so that we can divide by α):

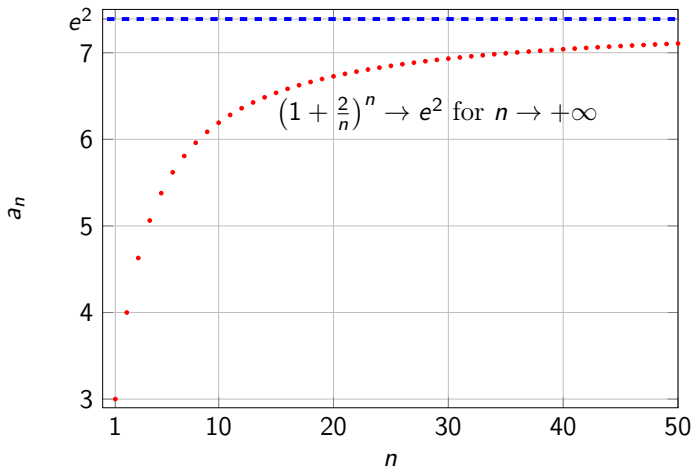
$$\left(1 + \frac{\alpha}{n}\right)^n = \left(1 + \frac{1}{\frac{n}{\alpha}}\right)^n = \left(\underbrace{\left(1 + \frac{1}{\frac{n}{\alpha}}\right)^{\frac{n}{\alpha}}}_{\rightarrow e}\right)^\alpha$$

Example: Computing limits with the sequence tending to e ($2/3$)

For example, with $\alpha = 2$ we have:

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{2}{n}\right)^n = e^2$$

Example: Computing limits with the sequence tending to e (3/3)



The number e :

- ▶ The sequence $\left(1 + \frac{1}{n}\right)^n$ is non-decreasing and bounded, hence convergent (monotone sequence theorem)
- ▶ Its limit is denoted by e (Euler's number), $e = 2.71828\dots$
- ▶ For every sequence $c_n \rightarrow \pm\infty$: $\left(1 + \frac{1}{c_n}\right)^{c_n} \rightarrow e$ and $\left(1 - \frac{1}{c_n}\right)^{c_n} \rightarrow \frac{1}{e}$
- ▶ These limits allow us to solve many indeterminate forms 1^∞

Euler's number in finance



Introduction: Euler's number in finance

- ▶ In this section we see how the number e arises naturally in the computation of **compound interest**
- ▶ What happens to the capital if interest is paid more and more often?

- ▶ Suppose we own a capital of unit value and an annual interest rate t . If the interest is paid annually, after one year the capital owned will be

$$1 + t \cdot 1 = 1 + t$$

- ▶ If instead the interest is paid monthly, we will have
 - ▶ after the first month, a capital equal to

$$1 + \frac{t}{12} \cdot 1 = 1 + \underbrace{\frac{t}{12}}_{\alpha}$$

- after the second month, a capital equal to

$$\underbrace{1 + \frac{t}{12}}_{\alpha} + \frac{t}{12} \underbrace{\left(1 + \frac{t}{12}\right)}_{\alpha} = \underbrace{\left(1 + \frac{t}{12}\right)^2}_{\beta}$$

- after the third month, a capital equal to

$$\underbrace{\left(1 + \frac{t}{12}\right)^2}_{\beta} + \frac{t}{12} \underbrace{\left(1 + \frac{t}{12}\right)^2}_{\beta} = \left(1 + \frac{t}{12}\right)^3$$

- ▶ at the end of the year we will have a capital equal to

$$\left(1 + \frac{t}{12}\right)^{12}$$

- ▶ If the interest is computed every n -th of a year, at the end we will have a capital equal to

$$\left(1 + \frac{t}{n}\right)^n$$

- ▶ For $t = 1$ (a 100% return) we obtain exactly the sequence that defines e

Euler's number in finance

Hence, even if the interest is paid an infinite number of times per year, the capital does not grow to infinity but tends to e , since:

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = e$$

Summary

Compound interest:

- ▶ With annual rate t paid n times a year, a unit capital becomes $(1 + \frac{t}{n})^n$ after one year
- ▶ For $t = 1$ and $n \rightarrow +\infty$ the capital does not grow to infinity but tends to e

Summary

- ▶ $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = e$: the limit exists by the monotone sequence theorem
- ▶ The result holds for any divergent sequence c_n , also with the minus sign (limit $\frac{1}{e}$)
- ▶ It is the main tool for the indeterminate forms 1^∞
- ▶ In finance, e is the limit of the capital when interest is paid infinitely many times a year